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RESEARCH ARTICLE

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Sustainable process improvements: Evidence from intervention-based research

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Abstract

This research develops a methodology for making process improvements that can sustain over time. Working with caregivers at a large U.S. hospital over 3 years, we redesign a process for educating kidney transplant patients with instructions for post-surgical care. Adopting an intervention-based research (IBR) framework and based on our actions to overcome challenges in implementation and sustainment of the redesign, we revise the current understanding of organizational learning theory. Follow-up observations after our intervention show that the process improvements at the hospital are sustained. We supplement the IBR with quantitative analyses and provide evidence of improvements in health outcomes and satisfaction levels of patients associated with the redesign. These analyses are based on difference-in-difference estimations using data from transplant patients, including a control group from other transplant units. Overall, our research specifies a methodology for implementing sustainable process improvements, particularly in high interaction service environments such as healthcare delivery, and identifies refinements to organizational learning theory, especially for such environments.

KEYWORDS

difference-in-difference, healthcare operations, intervention-based research, organizational learning, process improvements

1 | INTRODUCTION

The goal of process improvements (PI) is to redesign work processes with the systematic involvement of frontline employees (Nair, 2006; Tucker, 2007). Research in several industries has shown that organizational initiatives for PI are associated with better firm performance (Jacobs, Swink, & Linderman, 2015; Zhang & Xia, 2013). Despite the abundance of such research, there is a shortage of studies on sustaining PI over time (Collins & Browning, 2019; Morrison, 2013). Sustaining PI is critical to prevent backsliding of performance in specific processes and to ensure continual engagement of frontline employees (De Treville & Antonakis, 2006; Tucker, Nembhard, & Edmondson, 2007).

In the context of healthcare, numerous studies have provided evidence of the relationship between PI and hospital outcomes (Ding, 2015; Dobrzykowski, McFadden, & Vonderembse, 2016). Sustaining PI is particularly challenging in high interaction service contexts, in which customers participate with employees in the service delivery systems (Chase & Apte, 2007). In the high interaction context of healthcare, heterogeneity of patients and caregivers (e.g., inpatient nurses, physicians, outpatient nurses, and social workers) and necessary participation of patients and their families in delivery of care compound the challenges for sustaining caregivers' adherence to processes, and their engagement in PI (Damali, Miller, Fredendall, Moore, & Dye, 2016; Tucker & Edmondson, 2003).

While the importance of sustaining PI in healthcare organizations is recognized by practitioners and academics alike (Gardner, Boyer, & Ward, 2017; Pronovost & Vohr, 2010; Staats, Dai, Hofmann, & Milkman, 2017; Tucker & Edmondson, 2003), how to achieve such sustainment remains elusive. Thus, our broad research question is: *How to sustain PI in the context of healthcare delivery?*

To address this question, we participated in and studied the redesign of a process for educating kidney transplant patients with instructions for post-surgical care. This process requires high patient-caregiver interaction and active patient participation, thus presenting greater potential for variability, uncertainty, and complexity (Gordon, Prohaska, Gallant, & Siminoff, 2009). Patients' needs and requirements differ based not only on medical conditions and treatment plans (Englesbe, Dimick, Fan, Baser, & Birkmeyer, 2009), but also on patients' attention levels and ability to understand caregiver instructions (Dahl, Engebretsen, Andersen, Urstad, & Wahl, 2019). Lapses in following instructions can result in added complications leading to readmissions, emergency visits, and loss of life (Dew et al., 2007). Within this empirical context, our research question is: *How to redesign a process for educating kidney transplant patients and for sustaining the improvements from the redesign over time?*

Our intervention for the process redesign consisted of working with 32 caregivers, including frontline care providers (nurses, physicians, and social workers), support staff, and administrators at a large teaching hospital in the United States over 3 years, 2013–2016. We participated in, observed, and analyzed the execution of the process redesign while applying insights from organizational learning theory (Argote & Miron-Spektor, 2011; Argyris & Schön, 1978). Our specific focus was to redesign a process to include standardization and adaptation (Adler, Goldoftas, & Levine, 1999; Ansari, Reinecke, & Spaan, 2014) in patient education for post-surgical care, and to ensure sustainment of these changes over time (Anand, Gray, & Siemsen, 2012; Turner & Rindova, 2012).

We describe our rich experience of the process redesign and synthesize the lessons learned by adopting an intervention-based research (IBR) framework (Oliva, 2019). While applying organizational learning theory in our context of process redesign aimed at sustainment of PI, we faced numerous challenges that warranted adjustments for putting the theory in practice. Our problem context and these challenges fit the description of complex and ill-structured problems (Simon, 1973), and our adjustments reflect updating or refining existing theories as applied to such problems under the IBR framework (Oliva, 2019).

Overall, our intervention shows how the dual needs for standardization and adaptation encountered in high interaction service environments (Berry Jaeker & Tucker, 2020; Catena, Dopson, & Holweg, 2020) can be successfully delivered by building appropriate control and flexibility in service delivery processes. This involves giving employees guidance on protocols that should be strictly adhered to and adjustments that can be made based on customer requirements and needs (Nissinboim & Naveh, 2018; Sousa & da Silveira, 2019). Trusting frontline employees with these responsibilities and giving them a combination of direction and autonomy, in turn, increases their engagement not only in executing day-to-day processes but also in systematically seeking further improvements to the processes. We also find the role of middle managers to be vital in sustaining PI over time. Our experience in successfully implementing the process redesign for sustainment shows the importance of carefully managing PI in healthcare.

We assess the impact of our process redesign using data from 702 transplant patients spread over the duration of the redesign and including a control group for comparison. Specifically, we use a difference-in-difference (DID) approach to assess the impact on two patient outcomes: 30-day readmission and patient satisfaction. 30-day readmission indicates whether a patient was admitted to any hospital within 30 days after discharge and is commonly used to assess the quality of care delivered in hospitals (Senot et al. 2016a). Patient satisfaction is measured using the "Hospital Consumer Assessment of Healthcare Providers and Systems" (HCAHPS) survey, which is administered by the Center for Medicare and Medicaid (CMS). HCAHPS is a widely recognized measure of patients' perceptions of their overall experience receiving care at US hospitals (Sharma, Chandrasekaran, Boyer, & McDermott, 2016).

Our analyses show that the redesigned process is associated with improved outcomes over time. For the immediate term, the likelihood of 30-day readmission is 0.25 times lower and the HCAHPS score is about 8% higher for the treatment group as compared to the control group. For the longer term, a more than 1-year post-redesign assessment shows evidence of sustained use of the redesigned work process. Furthermore, we see continuing evidence to date (4 years after the redesign) of sustaining PI in the form of daily problem solving through huddles. These huddles involve and engage frontline employees and middle management in maintaining the patient education process and in considering ideas for further improvements of this and other processes.

The rest of the paper is structured as follows. In the section following this introduction, we relate our study to existing research in healthcare PI and organizational learning theory. Next, in Section 3, we present details of our empirical context of a high interaction service environment, and our intervention including its stages from first contact with caregivers to our exit after 3 years. In Section 4, we use an IBR framework, applying organizational learning theory in the redesign. Specifically, we apply existing knowledge of operational and change routines (Anand, Ward, Tatikonda, & Schilling, 2009; Nelson & Winter, 1982) and develop refinements based on our experience. Section 5 presents the results of our quantitative analyses for assessing the impacts of the redesign on patient outcomes. Section 6 concludes the paper with a discussion of how the refinements that we develop offer new perspectives for applying organizational learning theory to the sustainment of PI.

2 | SUSTAINMENT OF PI AND ORGANIZATIONAL LEARNING THEORY

The objective of this research is to implement sustainable PI in the high interaction service context of healthcare delivery. There is considerable research that associates organizational initiatives for PI in healthcare with performance; see Song and Tucker (2016) for a recent review of this literature. Only a few recent studies in healthcare operations management, listed in Table 1, and briefly reviewed here, have focused on the impact of a process change on the outcomes of the process. The association of the change with such outcomes provides a more direct test of the efficacy of the change as compared to association with organizational level outcomes, which represent higher-order effects.

In classifying the studies on process changes in healthcare, we first observed whether the researchers followed the implementation of a change concurrently or studied it retrospectively. Embedding researchers in the implementation of a process change gives the ability to more closely study the challenges faced and the measures taken to combat them. Second, we noted whether the outcomes of changes in these studies referred to patient outcomes, which are more closely related to process changes than overall hospital performance measures and assessments of caregiver adoption. Third, we included in our classification, information on whether the study covered sustainment of the results of the process changes over some period.

Using an existing healthcare system data set, Bavafa et al. (2018) observed that e-visits were associated with more office visits, had mixed results on phone visits and patient health outcomes, and were linked to a reduction in caregivers' accepting new patients. This research was based on a one-time instituted change and did not include any involvement of caregivers in the implementation of the change. In another study, employing existing data from two emergency departments, one of which switched from private to public disclosing of relative performance feedback (RPF) for physicians, Song et al. (2017) found the change to be associated with increase in productivity of physicians. The increase in productivity took a few months to become evident but, once realized, was sustained for about a year. While this study focused on intangible incentives for physicians, it similarly excluded information on their involvement in the process change. The initial dip in performance of the process outcome before realization of improvement showed the importance of assessing outcomes over an extended period. Staats et al. (2017) used data from hospital units to assess the effects of a deployment followed by discontinuation of an electronic monitoring system for compliance with handwashing instructions. Their results showed an increase in compliance associated with the deployment of

TABLE 1 Research on process change in healthcare operations management

Reference	Process change	Planned intervention	Patient outcomes	Test for sustainment
Bavafa, Hitt, & Terwiesch, 2018	Evaluate the impact of e-visits on patients' visit frequency, phone visits, and health	×	✓	×
Song, Tucker, Murrell, & Vinson, 2017	Compare the effectiveness of private and public disclosures of relative performance feedback	×	×	✓
Staats et al., 2017	Assess the impact of electronic monitoring on healthcare providers' compliance for handwashing	×	×	✓
Tucker & Singer, 2015	Evaluate the impact of management by walking around (MBWA) program in hospitals	✓	×	×

monitoring, followed by a gradual decline after its discontinuation. The inclusion of time over which they followed process compliance revealed this deterioration of results.

In all three studies summarized here, the researchers retrospectively analyzed the impact of changes to healthcare processes, and as such did not have the benefit of following the changes that were made with the involvement of frontline employees, which is an important aspect for effectively making and sustaining PI. Unlike with the previous three described papers, Tucker and Singer (2015) did use a planned intervention approach, subjecting hospitals to the treatment of a management by walking around (MBWA) improvement program that was developed based on multiple theoretical perspectives. Overall, they found MBWA implementations to have a negative impact on quality improvement efforts. However, the research did not report on any follow-up assessment after a period that may have been necessary to overcome that initial dip in performance.

Distinguishing from these studies, we implement a process redesign, applying theory, and assess the sustainment of results as well as offer lessons from putting the theory in practice (Edmondson, 1996; Holmström, Ketokivi, & Hameri, 2009). Our research is unique in that we (a) intervene in a process redesign, (b) focus on sustainment of PI in the organization, (c) offer refinements to theoretical concepts based on our experience, and (d) assess the impact of the redesign on dual outcomes of process effectiveness and customer satisfaction. Although our empirical context is healthcare, our research and its implications relate to sustainment of PI in all high interaction service environments.

We apply organizational learning theory (Argote & Miron-Spektor, 2011; Argyris & Schön, 1978) to implement PI in a healthcare context. Since Hayes, Wheelwright, and Clark (1988) identified the power of using the knowledge of employees to improve operations, and Leonard-Barton (1992) introduced the idea of shop floors as learning laboratories, several researchers have applied organizational learning theory to study PI in manufacturing and product development (Choo, Linderman, & Schroeder, 2007; Jayaram et al., 2010; MacDuffie, 1997; Mukherjee et al., 1998). In healthcare operations, involving frontline employees who deliver the services in the redesign of processes is especially helpful to increase their confidence in those processes (Nembhard & Tucker, 2011; Tucker et al., 2007) and to facilitate their use of standardization and adaptation in customer-facing processes (Berry Jaeker & Tucker, 2020; Catena et al., 2020).

Organizational learning theory has been used to study the effects of accumulated experience (Kc, Staats, & Gino, 2013; Pisano, Bohmer, & Edmondson, 2001) and initiatives for PI (Chandrasekaran, Senot, & Boyer, 2012; Ding, 2014; Roth, Tucker, Venkataraman, & Chilingerian, 2019; Song & Tucker, 2016) in healthcare. While the theory has been applied to explain the immediate results obtained from PI, it has been unable to address the issue of sustainment of results (Morrison, 2013; Staats et al., 2017). We address this limitation by working with caregivers to redesign a process while also aiming to maintain recurring cycles of organizational learning in practice (Anand, Ward, & Tatikonda, 2010; Nonaka & Von Krogh, 2009). We put in place a system for learning how to learn (Argyris & Schön, 1978) that goes beyond reactionary process changes (Tucker, Zheng, Gardner, & Bohn, 2020) and involves employees doing the work in redesigning how to do the work (Spear & Bowen, 1999).

In specific, we help caregivers establish standardized ways of doing their own work, called operational routines in organizational learning parlance (Nelson & Winter, 1982). The design of these routines includes adaptation-in-use known in theory as flexibility in the performative aspect of standardized routines (Feldman & Pentland, 2003). These routines and their outcomes serve as the baselines when implementing change routines based on results obtained from using the operational routines (Anand et al., 2009). Our intervention includes routines for root-cause analyses and for sustaining the search for improvements (Desai, 2020).

Our study uses an IBR approach (Edmondson, 1996; Oliva, 2019). It starts with understanding a complex problem in its empirical context and applying an existing theory to that context (Groop, Ketokivi, Gupta, & Holmström, 2017; Simon, 1973). Based on the lessons learned when addressing the problem, we make refinements to existing perspectives of the theory (Argyris, 1996). In Section 4, we map our process redesign and theory refinements to the components of the IBR framework (Oliva, 2019).

3 | CONTEXT AND METHODS

Our research site, which we call Alpha, is a large multispecialty teaching hospital located in the United States with roughly 1,500 beds, and 45,000 patient admissions per year. One of the critical problems for any hospital is to ensure patients have a good understanding of their own care after undergoing surgery, especially after transplants (Dahl et al., 2019). The Joint Commission for Accreditation of Healthcare Organizations has identified failure to render proper patient education for post-surgical care as a major cause of preventable readmissions (Regalbuto, Maurer, Chapel, Mendez, & Shaffer, 2014). Alpha was experiencing issues related to its processes for delivery of education to transplant patients for post-surgical care. Different units in the hospital had, in the past, deployed efforts for PI and redesigned care delivery processes including streamlining patient education for post-surgery care, only to find that the results of those attempts could not be sustained. Some members of the top management team at Alpha had approached one of the coauthors of this paper offering the opportunity to work on a redesign of the education process for transplant patients with the explicit aim of improving it in a way that sustains the changes and improvements. Another critical aspect of this involvement was that, if successful, this redesign would serve as a template for the hospital's renewal of an organization-wide initiative for PI that would include further such process redesigns.

Based on subsequent discussions and explorations of several units at the hospital, the academic research team and the caregiver research team jointly selected the kidney transplant unit for implementing the process redesign. This unit conducted approximately 200 kidney transplants per year and had an average 30-day readmission rate of 35%. Figure 1 shows an overview of the patient education process for post-surgical care before the redesign of the process, and points to some of the problems that were readily apparent. Some of them include variation in post-surgical care instructions, lack of proper handoffs between inpatient and outpatient caregivers, rushing through the instructions without assessing patient learning, and inconsistencies and errors in different patient education aids.

We present the timeline of our intervention for the process *redesign*, including *pre-redesign* and *post-redesign*, in Figure 2. The hospital representatives on our project team consisted of the chief quality and patient safety officer, physician director, and nursing director for the hospital's kidney transplant unit. Further, our research team interacted with all the personnel at the kidney transplant unit, comprising 15 inpatient nurses, 9 outpatient nurses, 2 nurse managers, and 12 transplant surgeons and nephrologists. We also interacted with the social workers, patient council staff, and information technology administrators to understand supporting features such as the patient handbook, medication lists, and electronic medical records (EMR) of patient education for post-surgical care. In addition, we solicited and got input from 15 kidney transplant patients. Details of our participation, observations, interviews, and surveys, which were conducted between January 2013 and April 2016, are presented in Table 2.

From January 2013 to June 2014, in the *pre-redesign period* indicated in Figure 2, the research team observed the existing patient education process and interacted with caregivers and patients to gather information. As part of this discovery phase, teams of two researchers shadowed caregivers while they delivered post-surgical care education for 15 patients. The researchers took detailed notes

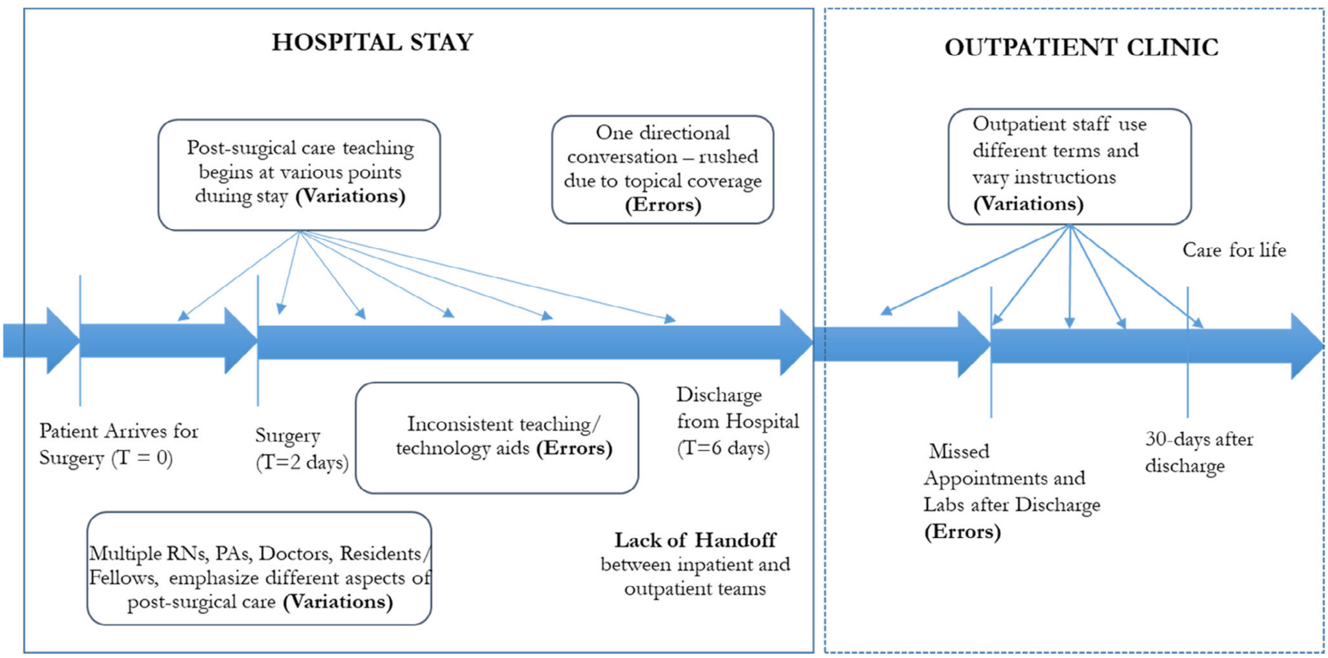

FIGURE 1 Education of kidney transplant patients prior to process redesign [Color figure can be viewed at wileyonlinelibrary.com]

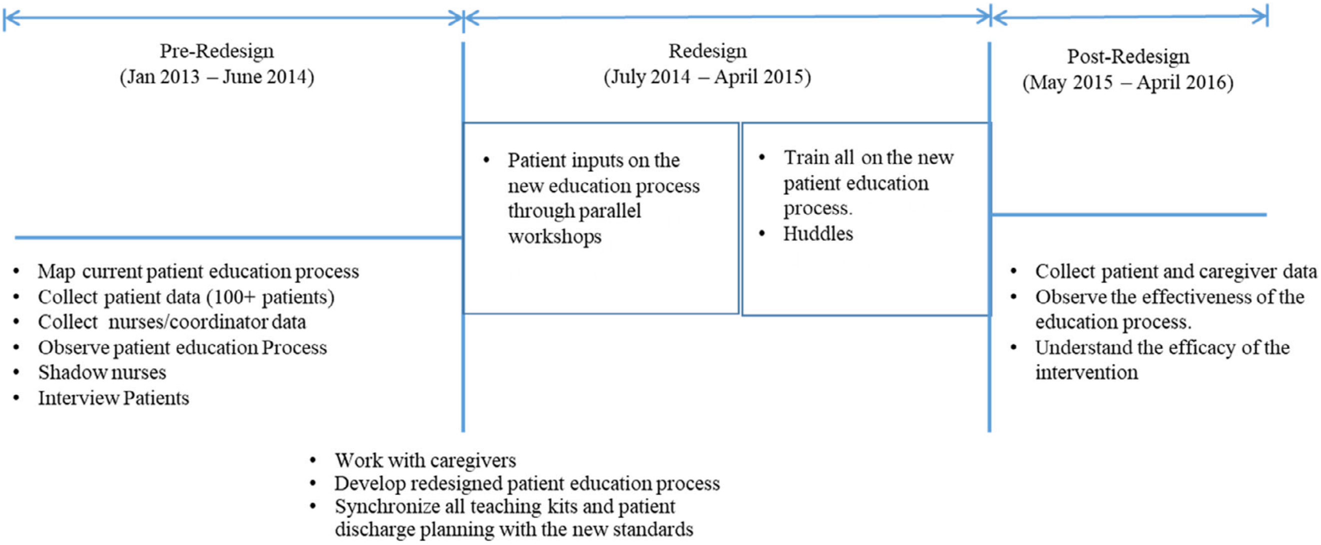

FIGURE 2 Implementation of the redesign for the patient education process [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 2 Types of data collected at each stage

Pre-redesign	Redesign	Post-redesign
• Field observations including caregiver shadowing and patient interviews • Survey data from 87 patients • Process maps • Post-surgical care instruction books and videos • Patient quality of care data • HCAHPS data • Patient clinical data	• Workshop data • Standard work process steps • Caregiver interviews • Patient group interviews • Patient quality of care data • HCAHPS data • Patient clinical data • Teach-back training minutes	• Field observations including caregiver shadowing and patient interviews • Survey data from 45 patients • Weekly huddle reports • Weekly nurse manager meeting reports • Patient quality of care data • HCAHPS data • Patient clinical data

TABLE 3 Components of qualitative data collection

34 interviews	
Respondents:	
	Chief quality and patient safety officer
	Medical transplant director
	Chief surgeon
	Nursing transplant director
	15 inpatient nurses
	9 outpatient nurses
	Social worker
	IT specialists
	3 patient council representatives
	Clinical psychologist
6 workshops (3 × 90 min, 3 × 180 min)	
Respondents:	
	Medical director
	Nursing director
	Inpatient and outpatient nurses
	Social workers
	IT specialists
	Clinical psychologist
2 patient focus groups (90 min each)	
Respondents	
	Former transplant patients (6 in each)

about the content and process of the education, focusing on engagement of patients and their families, and the use of educational materials by the caregivers. The observations from these visits were first individually transcribed immediately after the visits and then shared between the two researchers within 2 days to create a combined complete narrative. Next, we interviewed 34 caregivers listed in Table 3 to get their descriptions of the patient education process. Each of these interviews lasted about an hour and included topics like handoffs during the transition of care, roles of patients and their families during education, and challenges they faced related to the content and delivery of education. These interviews were transcribed, and the information collated.

The notes from the shadowing of education for the 15 patients and the transcription of 34 caregiver interviews served as input for six workshops with caregivers conducted during the *redesign period* from July 2014 to April 2015. These workshops were used to report to the caregivers our findings from the patient-education shadowing and to discuss areas in which there was variation within and between providers in the content and delivery of patient education. We used a workshop format to involve as many caregivers as possible at a time based on availabilities and to concentrate, in each workshop, on selected aspects such as patient handbook, medication lists, and electronic medical records (EMR) in seeking out improvements to the patient education process. In two of these workshops, we conducted value stream mapping (Rother & Shook, 2003), which included drawing the then-current state of the entire process of inpatient and outpatient care, including patient education, followed by drawing and agreeing upon a map of an envisioned future state.

In parallel with the caregiver workshops, we conducted two focus groups, each consisting of six patients, who had received transplants within the past 6 months, to get their inputs on the content and delivery of the education. These groups were diverse in terms of age, gender, donor type (cadaveric kidney versus living donor), pre-existing conditions, socioeconomic status, and education level of patients. Family members providing patient care accompanied most patients and provided additional input. The focus groups were conducted with the help of a facilitator specializing in the healthcare context. The questions used to structure the focus groups are listed in Appendix A. The 12 patients in the focus groups also responded to a survey related to their learning styles that was constructed based on an existing scale (Kolb, 1981). These questions are listed in Appendix B. Further, independent from these focus groups, we collected quantitative and qualitative data using a structured questionnaire and open-ended questions from 87 kidney transplant patients while they were at the hospital. This study was used to assess the relationship between overall care quality and patient outcomes.

In the *post-redesign period* (May 2015–April 2016), frontline nurses designed and implemented a system of problem-solving through daily (inpatient) and weekly (outpatient) huddles. This with done with input from our research team, including participation from middle management, and in agreement with top management. These huddles are short standup meetings (about 10 min) that are used to address day-to-day problems, share best practices, identify areas for PI, and track efforts for PI (Provost, Lanham, Leykum, McDaniel Jr, & Pugh, 2015).

4 | PROBLEM SITUATION (S), METHODOLOGY (M), AND THEORY REFINEMENTS (T)

The redesigned process that was the outcome of our intervention, engaging caregivers to improve patient education for post-surgical care after kidney transplants, is shown in Figure 3. We describe the redesign by adopting the mode of "interventions as a source of theory" (Oliva, 2019; p. 714), focusing on a real-world problem situation (S), which we tried to solve through a methodology (M) that we developed while refining an existing theory (T). Our problem situation (S) is implementing sustainable PI through redesign of a patient education process for post-surgical care, and we use organizational learning theory (T) with refinements generated while addressing the problem situation (S). The methodology (M) consists of working with frontline care providers and patients to understand the challenges faced in redesigning the patient education process for post-surgical care in a way that the changes are sustained, and developing and testing generalizable countermeasures to combat these challenges, resulting in refinements to the theory (T).

Our application of the IBR framework is consistent with abductive reasoning, trying to understand the challenges faced when designing PI for sustainment, followed by context-theory-context iterations, and new approaches to address those challenges (Chandrasekaran, de Treville, & Browning, 2020). In applying organizational learning theory to the problem context, we encountered challenges that questioned some of our current understanding of the theory. These challenges have been alluded to in the extant literature, but there have been few insights on how to overcome them. Thus, in conjunction with caregivers and patients, we developed and incorporated specific directions, which we call countermeasures, on how teams working on PI can overcome the challenges. We describe our intervention for the process redesign while mainly focusing on the four specific challenges we faced from starting to completion, and the countermeasures. These are summarized in Table 4.

4.1 | Initiation of redesign and engagement in PI

Through the early stages of our involvement, surgeons and nurses at the hospital showed little interest in even considering a redesign of the patient education process or efforts for PI in general. This was evident from our interviews with the caregivers. A nurse manager noted, "Our staff are already doing phenomenal work in terms of

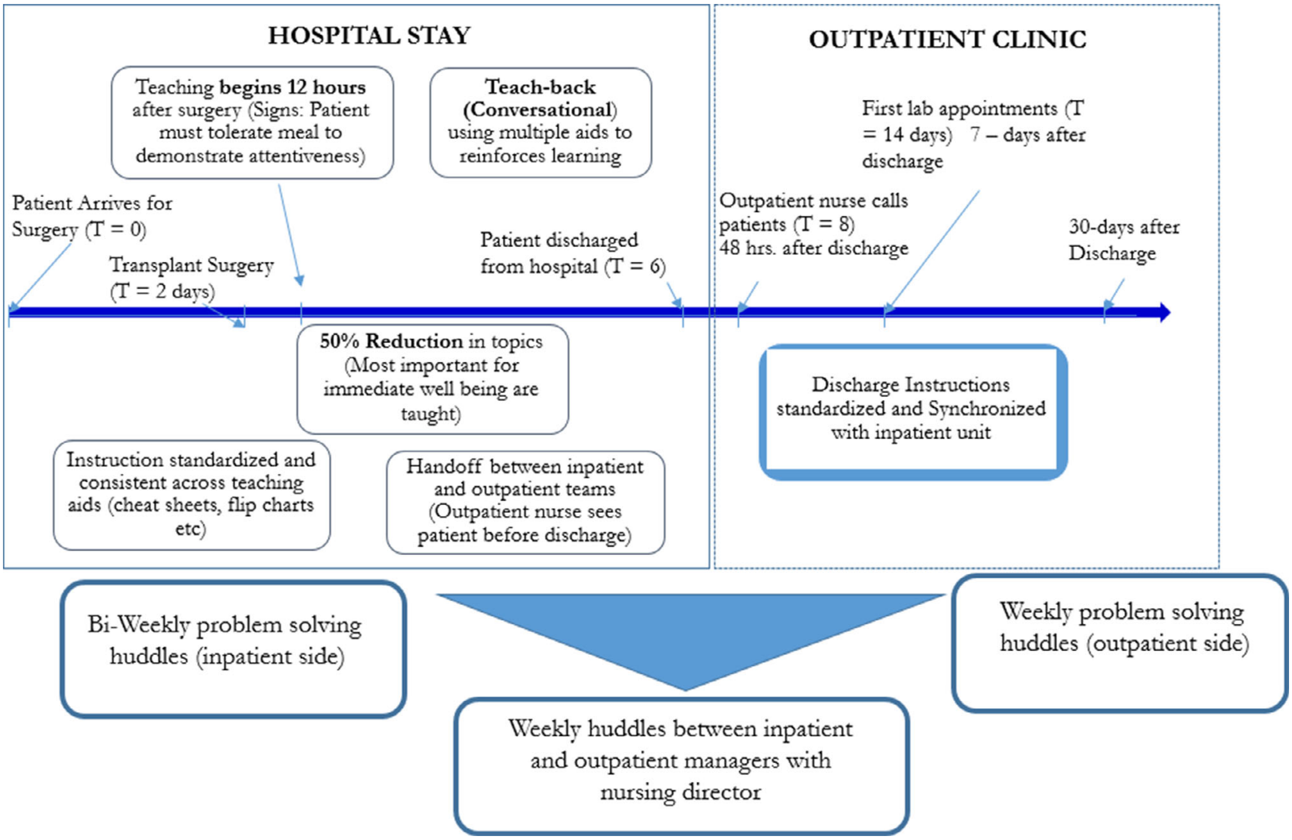

FIGURE 3 Patient education process for kidney transplant patients after the redesign [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 4 Challenges for sustaining PI and associated countermeasures

Challenges	Related literature	Current measures	Countermeasures/how to?
Initiation of redesign	Gowen III, Mcfadden, Hoobler, and Tallon (2006), Repenning and Sterman (2001)	Adoption of best practices	Self-diagnosis of need for change
Initial performance deterioration	Adler and Clark (1991), Nembhard and Tucker (2011)	Inevitability of tradeoffs in the short term	Frontline employee participation in designing process features
Adaptation in use	Ansari et al. (2014), Tucker (2007)	Customer specific employee autonomy	Customer input for redesign discrete from frontline employees
Sustainment of redesign and PI	Spear and Bowen (1999), Staats et al. (2017)	Employee engagement	Systematic involvement of middle management

educating the patients. In fact, many of our peers come to our transplant center to learn from us. We don't think there is an issue here." There was skepticism among the frontline employees about the need for even assessing the current process, let alone the need for its redesign. Even the surgical team was not persuaded by past research that proclaimed the benefits to be gained by efforts for PI. In the words of the hospital's chief surgeon, "You know, based on my previous work at a different medical center, I can tell that processes cannot be compared across sites. So, it is hard to transfer findings from another medical center to us. We are one of the largest ones in the world with a lot of complexity and bureaucracy. Issues that exist elsewhere may not exist here and vice versa." The first-line supervisors and middle management employees believed that their processes could not benefit from applying something that had worked in other hospitals and so did not see the value in efforts for PI. Our academic research team realized that name-dropping organizations such as Toyota, Cleveland Clinic, and Thedacare, and relating their success stories, or benchmarking processes with these organizations, only compounded the issue of employee resistance.

Research on organizational learning theory has recognized that efforts for PI take time, and keeping employees engaged in these efforts can be challenging (Gowen III et al., 2006). On the one hand, employees can be resistant to the idea of change (Kirkman & Shapiro, 2001). On the other hand, employees can be initially interested in making changes, but their interests may drop off, especially if there are significant efforts required with little or no benefit obtained in a short period of time (Nembhard & Edmondson, 2006; Siemsen, Balasubramanian, & Roth, 2007). Sustaining engagement of employees is critical for efforts toward PI, given that it takes time and effort to make changes, and to institutionalize and maintain them. As Repenning and Sterman (2001) noted, “You can't buy a turnkey Six Sigma quality program. Rather [successful implementation requires that] it must be developed from within” (p. 65).” In healthcare, top management support has been shown as one route for engaging employees, yet it has not proved to be enough (Chandrasekaran & Toussaint, 2019).

To address this challenge, we convinced the caregiving team at the kidney transplant unit, including the physicians, nurses, social workers, and information technology specialists, to participate in a discovery phase designed to study the existing process. Our statistical analysis of quantitative data coupled with the results of our qualitative analysis from interviews and process shadowing indicated that high variation in the existing patient education process was associated with greater patient anxiety levels. These findings, published in a healthcare journal (Chandrasekaran et al., 2016), generated awareness among the caregivers in the transplant unit of the need to redesign the process.

This discovery phase helped overcome employee skepticism about the value of attempting a redesign, and generated excitement for PI in general. Having participated in the examination of the process and upon seeing the results from their own context, caregivers were convinced of the need for rationalizing the process for patient education. Showing the opportunities that exist in their own context through a pre-redesign discovery study appears to be effective in combatting the challenge of engaging employees in PI, especially in a high interaction environment. This countermeasure identifies a refinement to organizational learning theory for addressing the challenge of engaging frontline employees in PI.

4.2 | Initial performance deterioration

Even efforts for PI that are successful often result in an initial decline in performance before producing desired results. For example, in their study of organizational learning in two electronic equipment manufacturers, Adler and Clark (1991) found that systematic learning efforts initially reduced productivity. Similarly, studying PI in automobile plants, Netland and Ferdows (2016) found that plant productivity worsened initially before improving. Nembhard and Tucker (2011) studied deliberate learning activities (DLAs) used for improving existing work practices in hospitals and found evidence of an initial dip in performance before turning around. Nembhard and Tucker's (2011) context is close to our context, and the authors of that paper called for future research to investigate impacts of DLAs (efforts for PIs) over a longer duration and based on in-depth observation.

One of the reasons for the initial decline in performance is the increased possibility of unplanned variations that may be introduced from the newly redesigned processes (Nembhard & Tucker, 2011). Another reason is the increase in cognitive workload when transitioning from existing processes to novel ones (Colligan, Potts, Finn, & Sinkin, 2015). Regardless, such decline in performance poses a challenge for the idea of PI in organizations. Employees get dejected and, after multiple tries, may dismiss such efforts as management fads (Repenning & Sterman, 2002). Even when forced to adopt the redesigned processes, employees may not be completely sold on the adoption, and may therefore not fully realize the potential improvements offered by the improved processes (Jasperson, Carter, & Zmud, 2005).

Keeping employees engaged in PI can help keep them well informed about the thinking behind each aspect of the redesign and can lead to watchful use of the changing process in the early days of its use (Adler et al., 1999; Kolb, 1981). Our research team organized the participation of the healthcare team in the *redesign period* from July 2014 to April 2015 as described in Section 3. We were able to maintain the engagement of the caregivers in the improvement effort, as indicated by this observation by an inpatient-nursing manager, “This process of breaking down post-surgery care instructions was an eye-opener. We were able to visualize the inconsistencies that are inadvertently created from our actions. Most of the time, the instructions and the patient education guidelines were passed on to us and we had no clear input on the process of delivering them.”

During the caregiver workshops conducted in the *redesign period* described in Section 3, participants noted that patients were overwhelmed by the amount of instructions. Several of these instructions included in the education (e.g., on topics such as gardening, dental health, and eye appointments) were non-critical to patients' immediate well-being and, thus, could more effectively be given at a later point of time by the outpatient team. This would distribute the information and make communication timelier, enabling better absorption and recall of the information among patients, leading to better adherence to patient self-care guidelines. Participants in the workshops also recognized the lack of consistency among caregiving team members in when and how education was delivered to patients, further pointing to the benefits that could be gained from standardization of these aspects.

As an example of the insights gained through these workshops, consider the evidence-based standard requiring kidney transplant recipients to drink at least three liters of water every day (Gordon et al., 2009). Throughout shadowing of patient education, conducted in the *pre-redesign period*, the research team noted variations in how the caregiving team delivered this specific instruction. While one nurse recommended the patient drink "a lot of fluids," another nurse suggested "two liters of water," while a third suggested "100 ounces of water." The research team also observed varying levels of empathy in caregiver–patient interactions. Caregivers in some instances rushed through patient education due to a variety of circumstances (e.g., in one instance, a nurse delivering patient education was called away to assist for a biopsy). We discussed this and similar additional inconsistencies in practices in the workshops. With the aim of eliminating such inconsistencies and incorporating patient perceptions, the caregiving team took on the charge of redesigning the patient education process and incorporating feedback from patient focus groups (described in Section 3) conducted in parallel with the caregiver workshops.

With the redesigned process, the caregiving team adopted a two-part patient-education approach (see Table 5), with inpatient nurses delivering the most essential education for post-surgical care during the hospital stay (Part I Instructions), and outpatient nurses delivering the rest of the education for post-surgical care after discharge from the hospital (Part II Instructions) approximately 48 hr after discharge. An important part of the redesigned process is that outpatient nurses are now required to meet the patient and the inpatient coordinators prior to discharge to initiate a formal handoff process. This is aimed at reducing errors that can occur because of unmindful handoffs (LeBaron, Christianson, Garrett, & Ilan, 2016).

Overall, while employee engagement in the redesign resulted in specific insights about the patient education process described here, the main insight that is new and generalizable from our study is about avoiding the initial dip in performance. We found that because of their active engagement, the frontline caregivers were indeed able to better take care of the teething problems, separating such problems from ones that needed further investigation for improving the process. Participating in redesigning the process also allowed them to build in some adaptation-opportunities, along with standardized practices, which turned out to be useful in mitigating the effects of unplanned variations. These aspects of our implementation resulted in minimizing initial disruptions to the process as well as in deriving lessons from the initial use of the redesigned process. Thus, frontline employee involvement from the start of the process redesign appears to be an effective countermeasure for avoiding the tradeoff between customer satisfaction and process effectiveness, which was accepted as *fait*

TABLE 5 Post-surgery care instructions

Major topics pre-redesign	Major topics post-redesign
• Infection prevention • Symptoms of rejection • Labs • Dental care • Eye care • Vaccinations • Sports and recreation activities • Lifestyle changes • Going back to work • Treatment of complications • When to call • Medications • Vital signs • Gardening • OTC meds • Follow-up appointments • Fluid intake • Activity progressions (e.g., pregnancy, sexual activity, special activity) • Pet care • Holiday schedules for labs • Wound care • Going out to public places • Lifting Instructions • Emergency contact (e.g., 911, primary care) • Smoking and drinking • Use of EMR/patient portal	*Part I (during hospital stay) by inpatient nurses* Starts within 24 hr post-transplant and no later than 32 hr; additional conditions: RASS score = 0, companion present and patient tolerated first meal • Infection prevention • Vital signs and symptoms • Labs • When to call • Medications • Vital Signs • Fluid Intake • Emergency contact (e.g., 911, primary care) *Part II (after discharge) by outpatient nurses* Starts within 48 hr after discharge and continues for about 3 months (face-to-face as well as over the phone) • Medications • Follow-up appointments • Activities (e.g., pregnancy, sexual, social, gardening) • Vaccinations • Eye and dental care • Lifestyle change • Treatment of complications • Post-op care (wound care) • Emergency contact (e.g., 911, primary care) • OTC meds • Going back to work, public places • Pet care • Exercising and sports activities

accompli in past applications of organizational learning theory to PI.

4.3 | Adaptation in use

Based on the organizational learning perspective, processes should have adherence requirements that are meaningful so that their execution is not ad hoc (Johnson, Burgess, & Sethi, 2020); at the same time, processes should not be so inflexible that they fail to add value to their underlying objectives (Ansari et al., 2014). If processes are designed to be overly restrictive and do not allow requisite adaptability by the user, employees may not adopt such processes (Nissinboim & Naveh, 2018). The duality of standardization and adaptation can be addressed by designing a process with ostensive aspects that focus on the principle of the routine that must be adhered to but also including performative aspects of routines that provide the user some degree of flexibility in the use of the routine (Anand et al., 2012).

Developments around this perspective in the marketing literature have advocated incorporating the role of the customer in designing processes that combine standardization and adaptation (Auh, Menguc, Katsikeas, & Jung, 2019; Bendapudi & Leone, 2003). Applying this perspective to organizational learning theory, we sought input from patients to build in adaptation in use by frontline caregivers for standardized practices. Although the importance of incorporating customer voice into process design has been recognized (Catena et al., 2020), there is little insight on how to effectively do this in a healthcare setting. Especially given the active involvement of frontline employees in our context, the research team wanted to ensure we were getting unbiased and frank input from patients. The challenge for getting such patient input is illustrated by the following observation from a transplant patient, "After our discharge, the post-transplant coordinators (nurses) are extremely critical for our well-being. It is awkward to raise issues when they are in the same room as us."

Thus, we opted not to survey customers or conduct provider–customer focus on the involvement of caregivers. Instead, we conducted facilitator-led focus groups of patients separately but in parallel with the caregiver workshops (described in Section 3). This enabled an interchange of ideas across both forums while maintaining separation to protect the privacy of their comments. This combination helped consider, among other suggestions from patients, aspects that warranted either more standardization or more adaptability-in-use for caregivers in the design of practices.

The first patient focus group meeting was used to adjust the topics included in the two parts of the patient education work (i.e., the contents of Part I and II instructions shown in Table 5). Note that patient feedback was used primarily to inform sequencing, not for content, which was determined based on evidence-based best practices. Patients attending these meetings also pointed to the need for having multiple teaching styles to reinforce the learning. This was triangulated with data collected using the Kolb (1981) instrument for assessing individual learning styles. In the second focus group, patients discussed outcomes and instruction delivery aspects (e.g., use of visuals and teach-back standards) and helped resolve inconsistencies in the delivery. As an example, patients noted the confusion that arose when inpatient nurses used a medication's chemical name (e.g., "Cyclosporine") while outpatient nurses used its generic name (e.g., "Neoral"). We worked to resolve such issues through the caregiver workshops being conducted in the same period and then rolled out a training program for all the nurses on the delivery of new instructions. Table 6 gives illustrative examples of the changes in the patient education process that resulted from the redesign.

In general, having patient input in parallel provided insights that were not otherwise considered by the caregiving team. It also helped the caregiving team to build in some adaptation to accommodate variations. As an example, data from the focus groups suggested that some patients preferred visuals for education while others preferred detailed explanations through narratives. This information was then used by the caregiving team, which developed both visual and conversational teaching (teach-back) techniques. The results of the learning styles questionnaire used in the focus groups indicated the importance of several learning styles needing to be integrated into patient education—so we included several methods such as reflection, visual aids, teach-back, and quizzes before discharge. Having patient feedback through parallel workshops provided useful insights into improvement opportunities and enabled the caregivers to design in adaptability for reacting to the varying needs of the patients and other changes in the task environment. The topics for this education are shown in Table 5.

4.4 | Sustainability of redesign and engagement in PI

Adherence to established processes is challenging to sustain over time (Anand et al., 2012; Desai, 2020), especially in high interaction service settings (Ton & Huckman, 2008). Take, for example, Staats et al. (2017)'s

TABLE 6 Examples of pre- and post-redesign practices in patient education

Patient care issue	Pre-redesign	Post-redesign
Recommended water intake of 3 L every day	Variation in the quantity of water: e.g., "a lot of water", "2 L", "96 oz",…	Uniform adherence facilitated by a 1.5-L container given to patients at discharge, to be consumed twice daily
Signs and symptoms of rejections and infections: Fever greater than 101 F (oral), weight loss greater than 2 lbs in a day or 5 lbs in a week, swelling, decreased urine output, drainage from the wound, tenderness and swelling	Variation in the temperature, extent of weight loss, missing instructions on swelling and urine output, missed instructions on infections	Checklist with specific signs and symptoms of rejection vs. infection Rejection: Fever, weight loss, urine output, swelling Infection: Drainage from wound, tenderness
Medication & lab Instructions (labs 2/week for 3 months), Neoral medication (when to take—15 min before or 2 hr after labs)	Missed instructions on number of labs/week and why to do these labs. Frequent confusion of medication names (Neoral vs. cyclosporine, timing and frequency not always emphasized)	Standard work for caregivers to deliver these instructions Content—Proper name for medication Sequence—When is it appropriate (before or after labs) Timing—How soon (15 mins) or later (2 hr after lab) Outcome—What are the consequences of missing these medications
Who to call (911 vs. going to ED vs. calling nurse coordinator)	No consistent explanation on when to call 911 or go to ED vs. calling the transplant coordinator	Standard work for caregivers to deliver these instructions Call 911 or go to ED if chest pain, shortness of breath, or any other medical emergency Call transplant coordinator for lightheadedness, fever (101F oral or higher), nausea or vomiting

observation that even after caregivers established handwashing routines under monitoring, compliance rates dropped soon after monitoring was discontinued. Related to this challenge, sustained engagement in PI—getting employees excited about trying out more process redesigns for improvement—is also a challenge that is widely acknowledged as being difficult (Adler et al., 1999; Spear & Bowen, 1999).

We recognized through the experience of the design and implementation of the process redesign that middle managers (nursing managers) could play a critical role in sustaining the new process as well as the initiative for PI (Sitkin, 2020). However, some previous research had pointed out that as the conduit between frontline employees and top management, middle managers tend to have their contributions muddled in reconciling top- and bottom-rank views (Birken, 2019). The institutionalization of a system of huddles noted in Section 3 helped achieve the objective of defining responsibilities and creating a system and structure for involvement of middle managers in PI. The huddles also ensure that handoffs among employees are seamless (Prætorius & Hasle, 2019). Examples of topics discussed in these huddles are:

- Who are getting discharged—and their aptitude on post-surgical care instructions?
- Staffing levels in the unit for that day
- Hand-off process between inpatient and outpatient unit (e.g., has the outpatient coordinator met the patient and discussed Part 2 of the instructional delivery?)
- Improvements to teaching process
- Celebrations and other news

The nurse manager (middle management) from the inpatient unit had the following comment on the use of huddles, "At first, we were skeptical with the role of daily meetings, we have had this in the past and it failed to sustain. However, given the systematic effort of involving me and my staff from the beginning, I felt it was my role to motivate and keep this improvement process going on over time. We have great buy-in from our day and night shift nurses given their early involvement and they own it within the unit."

This system gained acceptance almost immediately, and outpatient and inpatient nurse managers (i.e., middle management) made this a part of their weekly and biweekly routines. They created a structure to facilitate the huddles (<10 min), monitored them, and offered problem-solving support. This gave rise to a weekly Monday meeting between the inpatient and outpatient manager that the nursing director (senior management) also attended. Among other things, managers shared information in these huddles on patients and the patient education process and updated each other on huddles in their respective units. The use of these huddles organically created a tiered management structure that now facilitates escalation of frontline issues to middle management and senior management as appropriate. Our subsequent observations show that this system is helping sustain adherence and exploration of PI in the redesigned patient education process. It has also propagated the practice of huddles in other areas of the hospital, thus helping sustain process-adherence and engagement in PI. We also noted that building in adaptation-in-use for processes, in response to the third challenge described earlier, helped with frontline employee engagement in the organizational initiative for PI.

The huddles that began in May 2015 were still being regularly conducted in the unit as of May 2019. We found the nurse managers to be pivotal in ensuring that frontline nurses had the necessary resources. The nurse managers also met with the nursing director (senior leadership) on a weekly basis to communicate information across the inpatient and outpatient areas.

Overall, our research demonstrates that PI can sustain over time even in a high interaction service environment such as healthcare delivery. The countermeasures for sustaining PI, which were designed inductively, working with the caregivers, also resulted in refinements to organizational learning theory as summarized in Table 4. Formalizing these refinements to the theory would require testing these countermeasures in different contexts, as well as comparing our unique adjustments to other ways for tackling the challenges in similar contexts (Groop et al., 2017).

5 | EFFICACY OF THE PROCESS REDESIGN

To legitimize the role of our intervention as a template for process redesign and organization wide initiatives for PI, it was critical to assess the efficacy of the redesign in addressing this problem. Thus, we examined the impact of redesigning the patient education process by comparing 30-day patient readmission rates before and after the redesign with respect to a control group. We also compared patients' perceptions of care in the kidney transplant unit before and after the implementation of the redesign with respect to the same control group. We added this performance dimension because, in healthcare contexts, especially those that involve high patient–caregiver interactions, patients' perceptions of care are not always compatible with their health outcomes (Berry Jaeker & Tucker, 2020; Kang, Shah, & Dowd, 2017). At the same time, patients who report higher satisfaction levels would better absorb and use information about post-surgery care, increasing their chances of better health outcomes (Nair, Nicolae, & Narasimhan, 2013; Sharma, Chandrasekaran, & Bendoly, 2020). Thus, we analyzed the impact of the process redesign by comparing before and after scores on the HCAHPS (Centers for Medicare and Medicaid Services (CMS), 2019) for the kidney transplant unit, and with respect to a control group.

During January 2013–April 2016, 571 patients underwent kidney transplants at Alpha. These patients constituted the treatment group for our difference-in-difference (DID) analysis (Imbens & Wooldridge, 2009). We compared this patient mix with the population of all kidney transplants conducted in the United States during the study period using data obtained from the National Kidney Foundation (https://www.kidney.org/). The average readmission rate for our sample was 40%, which is not significantly different from the national average ($p > .25$) during the study period. Differences in other demographic and donor-type statistics also were nonsignificant ($p > .30$), giving further reassurance of the representativeness of the sample.

We collected performance data from two other transplant units, heart and liver, within the hospital that together constituted our control sample for the DID analysis after adjusting for patient and other transplant characteristics. Patients undergoing heart and liver transplants were getting similar post-surgery care instructions as kidney transplant recipients. Their post-transplant education process, moreover, shared many commonalities with the process for kidney transplants. It is important to note that the heart and liver transplant units only experienced routine hospital-wide quality and safety efforts and did not make other changes such as our process redesign, during our period of study. We collected patient and process data for the 62 heart and 103 liver transplant patients conducted in the same period (January 2013–April 2016). Having a control sample from the same hospital, we believe, provides a strong test of the effectiveness of our process redesign. Appendix C contains descriptions of the variables used in our analyses, Appendix D gives the means and standard deviations, and Appendix E shows correlations.

The process redesign constituted our quasi-experimental treatment that resulted in an exogenous shock to the patient education process beginning in January 2014. We used a DID approach (Imbens & Wooldridge, 2009) to estimate the causal effects of the redesign in the patient-education process. This approach accounts for differences in baseline 30-day readmission rates and HCAHPS scores across the two transplant groups and mitigates the effects of any changes other than the process redesign. We checked for the necessary conditions including parallel trends before conducting a DID analysis. Appendix F gives the details of these tests.

5.1 | 30-day readmission

We investigated the post-redesign change in patient outcomes by examining whether a transplant patient got readmitted to the hospital within 30 days in the treatment and control groups. We estimated the following relationship at the patient level for comparing redesign (redesign or post-redesign) and pre-redesign readmissions:

$$\begin{aligned} logit\,(Readmission)_{i,j,t} &= \beta_0 \\ &+ \beta_1\, Type_{pij} + \beta_2\, Phase_{S = Redesign\ or\ Post} \\ &+ \beta_3\, Type_{pij} {*} Phase_{S = Redesign\ or\ Post} + \theta_{jt} + T_{\alpha j} + \delta X_{ijt}. \end{aligned}$$

In this equation, i indexes each patient, j indexes each transplant type, and t indexes time in year and quarter periods. $Type_{pij}$ captures the type of transplant patient (kidney, heart, or liver); $Phase_{S\ =\ Redesign\ or\ Post}$ represents periods of the redesign (*Redesign* or *Post-redesign*) compared to the *Pre-redesign period*. $Type_{pij} * Phase_{S\ =\ Redesign\ or\ Post}$ captures interaction between transplant type and time, while $T_{\alpha j}$ represents time-invariant transplant controls. $\underline{X}_{ijt}$ represents patient-level controls and θ_{jt} represents time-variant process controls for the transplant groups. More information on controls including the rationale for using them is presented in Appendix C. We used logistic regression with robust standard errors (Wooldridge, 2002) and interacted the treatment group dummy variable with the treatment periods to investigate readmission trends. As a verification check, we used the diff command in STATA 14 and controlled for all covariates shown in Appendix D; the results were consistent. The variance inflation factor (VIF) in all our models was well below the threshold value of 10 (Hair, Anderson, Babin, & Black, 2010).

Table 7 shows the results of this analysis. Models 1 and 2 show the likelihood of patient readmission in the *redesign* and *post-redesign periods* It is important to note that these regressions were run comparing the patient education process of the treatment group (kidney) with the control group (heart and liver) in the *redesign* and *post-redesign periods* with their pre-redesign characteristics, which explains different sample sizes for both these regressions (n = 489 patients for redesign and n = 528 for post-redesign). It is also important to note that some process controls (e.g., *Transition and Integration*), described in Appendix C, occurred post-redesign, and were therefore dropped in the redesign model (Model 1).

Results of Model 1 (Table 7) suggest that the patient's likelihood of readmission in the treatment group during the *redesign period* is no different from that in the control group and *pre-redesign period* ($b = -0.07$; $p > .10$). This suggests that the benefits of our process redesign can only be seen after the patient education process was fully implemented. Nevertheless, we also found no

TABLE 7 DID results for impact of process redesign on 30-day readmissions (Pre-Redesign is the base group)

	Redesign Model 1	Post-redesign Model 2
Redesign or post-redesign period	1.06 (0.68)*	−0.29 (0.61)
Kidney	−0.08 (0.41)	0.06 (0.41)
Redesign (or post- period) x kidney	−0.07 (0.52)	−1.39 (0.71)**
Patient controls		
Heart	−0.74 (0.50)*	−0.95 (0.48)**
Mortality	−0.65 (0.52)	−1.84 (1.52)
LOS	0.05 (0.02)**	0.07 (0.02)***
Age	0.00 (0.00)	0.00 (0.00)
Gender	0.55 (0.20)***	0.30 (0.20)
Donor type	−0.21 (0.22)	−0.26 (0.22)
White	0.23 (0.46)	−0.07 (0.45)
African American	0.30 (0.50)	−0.08 (0.48)
Graft health	0.56 (0.40)*	0.64 (0.33)**
Pre-existing condition	−0.09 (0.14)	−0.00 (0.14)
Process controls		
VBP	0.96 (0.57)**	1.82 (0.73)**
Transition	NA	1.69 (0.40)*
Integration	NA	−1.38(0.35)
Psuedo R^2	5.20%	5.52%
Wald χ^2	29.37**	35.32***
Obs.	489	528

Note: Year and Quarter dummy variables, added as controls, are not shown here. *$p < .10$; **$p < .05$; ***$p < .01$, robust standard errors in parentheses.

deterioration in performance in this period compared to the patients in the control group, as expected based on the prior literature (Nembhard & Tucker, 2011; Song et al., 2017). Readmission across both the treatment and control group, however, was worse than in the other periods. That is, the coefficient of *Redesign* is significantly associated with readmissions ($b = 1.06$, $p < .10$, odds ratio [OR] $= 2.89$), suggesting the likelihood of readmission within 30 days was significantly higher in this period.

Model 2 (Table 7) provides the results for the likelihood of readmission post-redesign. As seen from Model 2, the coefficient of the interaction term between redesign (post) and Kidney is negative and significantly associated with readmissions ($b = -1.39$, $p < .05$, OR $= 0.25$). This suggests that kidney transplant patients' likelihood of readmission within 30 days was lower post-redesign compared to the control group (heart and liver) and to the *pre-redesign period*. Specifically, post-redesign odds of readmission within 30 days were 0.25 times lower when compared to the control group and pre-treatment periods. The patient controls and time-relevant process controls remain consistent across both regression models, suggesting that their effects were consistent throughout the study period.

5.2 | Patient satisfaction

We accessed monthly HCAHPS data for both the kidney transplant unit (Unit A) and the liver[1] unit (Unit B) in the hospital. We estimated the following relationship between the process *redesign periods* and the overall satisfaction ratings at the unit level.

$$\begin{aligned} &\text{Overall Patient Satisfaction}_{i,t} = \beta_0 \\ &+ \beta_1\, Type_{pi} + \beta_2\, Phase_{S=Redesign\ or\ Post} \\ &+ \beta_3\, Type_{pi} {*} Phase_{S=Redesign\ or\ Post} + \theta_{jt} + T_{\alpha j} \end{aligned} \quad (2)$$

In this equation, like Equation (1), i indexes the transplant type and t indexes time in year and months periods. $Type_{pi}$ captures the type of transplant patient ($i = 1$ for kidney; $i = 0$ for liver); $Phase_{S\ =\ Redesign\ or\ Post}$ represents the periods of the process redesign (*Redesign* or *Post redesign*) compared to the *pre-redesign period*. $Type_{pi} * Phase_{S\ =\ Redesign\ or\ Post}$ captures transplant type and time interaction, while $T_{\alpha j}$ represents time-invariant transplant controls and θ_{jt} represents time-variant process controls for the transplant groups. We use robust standard errors (Wooldridge, 2002).

Table 8 gives the results of these analyses. As in our earlier analyses, we included all process-level controls (e.g., VBP, EMR, etc.), described in Appendix C, along with relevant time intervals when comparing the two units. Models 3 and 4 provide results for *redesign* and *post-redesign periods*, respectively.

Model 3 (Table 8) compares the trends between the baseline period and the *redesign period*. As seen from Model 3, the interaction term is negatively associated with HCAHPS score ($b = -0.11$, $p < .01$), suggesting a decline in HCAHPS performance for the kidney transplant unit between the redesign and baseline periods, when compared to a similar change in HCAHPS scores for the control group. Model 4 gives the HCAHPS results for the *post-redesign period*. As seen from the results for Model 4, the interaction term between Redesign (Post) and Kidney is positively associated with the overall patient satisfaction ratings ($b = 0.09$, $p < 0.05$). This suggests overall change in HCAHPS scores for the kidney unit between the post-redesign and baseline period was higher when compared to a similar change for the control group.

We verified the robustness of our empirical results for both sets of analyses, namely the 30-day readmission and patient satisfaction. The robustness checks comprised changing the time window for impact of process redesign from 12 months to 9 months, and repeating our main analyses examining only cadaveric kidney transplants as the treatment group, and comparing it with the heart and liver patient control group. The interpretation of our substantive results remained unchanged.

TABLE 8 DID results for impact of process redesign on HCAHPS scores

Pre-redesign is the base group	Redesign Model 3	Post-redesign Model 4
Redesign or post-redesign period	0.03 (0.04)	−0.06 (0.05)
Kidney	−0.07 (0.06)	−0.06 (0.08)
Redesign (or post-period) × Kidney	−0.11 (0.05)***	0.09 (0.04)**
Patient controls		
Number of transplants	0.07 (0.03)**	0.08 (0.04)**
Process controls		
VBP	0.07 (0.04)+	0.06 (0.04)+
Transition	NA	−0.03 (0.07)
Integration	NA	−0.10 (0.11)
Adjusted R^2	19.08%	28.73%
F	2.56**	3.38**
Observations	54	60

Note: Year and Month dummies are added as controls. *$p < .10$; **$p < .05$; ***$p < .01$, robust standard errors in parentheses.

6 | DISCUSSION

6.1 | Implications for theory

The purpose of this study was to develop a methodology for sustaining PI in high interaction service organizations such as hospitals. There is limited research in operations management and organizational science that has offered concrete recommendations on how to sustain PI. Toward this purpose, we participated in the redesign of an education process for kidney transplant patients for post-surgical care. The four challenges to sustaining PI that we identified, namely, initiating process redesign, initial performance deterioration, incorporating adaptation-in-use, and sustainability of redesign and engagement in PI, had been discussed in the extant literature without guidelines for overcoming them. Our research offers preliminary evidence on ways to overcome these challenges and sustain PI. We highlight four main contributions of this research.

First, our research finds that conducting a process discovery prior to considering process redesign assures employee buy-in for initiating PI. Although we had senior management support at the outset, the nurses and surgeons at the hospital were uncommitted to the process redesign. Initial interviews with nurses suggested that some surgeons were even dissuading them from participating. The results of our pre-redesign study, however, generated a shift in the attitudes of nurses and surgeons toward the redesign, leading to their enthusiastic participation. While this evolution, from skepticism to enthusiasm, is often alluded to as being warranted (Adler et al., 1999; Nembhard & Tucker, 2011; Staats et al., 2017; Sterman, Repenning, & Kofman, 1997), the extant research does not describe how the shift can be realized. Current understanding of operational and change routines from organizational learning theory (Desai, 2020) does not offer insights on how to overcome such skepticism. This critical gap is perhaps responsible for the failure to sustain organizational initiatives for PI across all types of settings including manufacturing. By demonstrating the effectiveness of a pre-redesign study and analysis on caregiver acceptance of the need for process redesign, we add specificity that can help organizations better initiate process redesign and efforts for PI that are sustained over time.

Second, our research offers insights on implementing process redesign in high interaction service settings. We included patient input in the redesign of the process in a parallel fashion. Although previous studies on process changes have advocated for customizing service process delivery (Boone, Ganeshan, & Hicks, 2008), they shed little light on how to incorporate customer input during design and development of processes. Studies suggest surveying customers to gauge their perceptions of the process and using them for the design (Bendapudi & Leone, 2003; Damali et al., 2016). Others recommend using service provider–customer focus groups to simultaneously consider their viewpoints (Gittell, 2003). Neither of these strategies are effective for capturing simultaneous inputs from patients and providers. Although patient input can be useful to understand modes of delivery of education for post-surgical care, capturing it in the end would not have been useful for redesigning the process. Our research, in contrast, delineates a new way for incorporating patient feedback in improving healthcare delivery. Hospitals find themselves continually redesigning service delivery in response to reimbursement changes (e.g., VBP requirements—Centers for Medicare and Medicaid Services (CMS), 2019) or new care delivery models (e.g., Accountable Care Organizations—Shortell, Wu, Lewis, Colla, & Fisher, 2014; Population Health Models—Moffatt-Bruce, McAlearney, Aldrich, Latimer, & Funai, 2014). Our findings suggest that using the caregiving team's evidence-based standards for process change and, in parallel, incorporating patient-centric features based on the feedback from patients can be an effective approach for redesign. The initially sequential then separate but simultaneous collection of caregiver and patient viewpoints can be useful in synthesizing information and counterbalancing process features. This combines the benefit of standardization with the ability to adapt certain features based on the patient needs and preferences. Ultimately, this can help achieve better patient health outcomes and higher customer satisfaction.

Third, we offer insights on how to maintain organizational initiatives for PI and create a mindset of continuous improvement. Evidence on how to sustain PI is lacking, even outside healthcare (Argote & Hora, 2017; Choo et al., 2007). In our research, success in sustaining the initiative introduced through our intervention can be attributed to the midlevel management's continuous engagement in huddles described earlier. This middle-up-down method also helps frontline nurses and senior leadership connect in a timely and sustained manner. Nurse Managers from the heart and liver transplant control group noted that such coordination of care in their units was, in contrast, mostly ad hoc and non-routine. Limited empirical evidence exists on the role of middle management in sustaining PI (Birken, 2019; Fryer, Tucker, & Singer, 2018). In practice, middle managers are often labeled as the "muddle in the middle" and are viewed as being adversarial to PI (Burgess & Currie, 2013). In our research, we found these managers to be critical elements in sustaining PI following process redesign. Expecting senior management (e.g., medical and nursing directors) to exhibit the same level of effort and commitment to daily problem solving was implausible given the large size of the hospital. Instead, inpatient and outpatient nursing managers took the lead—not by leading huddles but by supporting frontline nurses. They also acted as liaisons between frontline nurses and medical and nursing directors, ensuring necessary resources for sustaining standardized practices.

Our fourth contribution comprises a demonstration of the IBR approach for studying ill-structured problems such as sustaining PI (Chandrasekaran et al., 2020). Thus far, knowledge about sustaining PI had been derived from case studies that predominantly used inductive reasoning (e.g., Adler et al., 1999; Eisenhardt, Furr, & Bingham, 2010) or from pre- and post-implementation comparison studies that predominantly used deductive reasoning (e.g., Choo, Nag, & Xia, 2015; Nembhard & Tucker, 2011). While both methods have their merits, they do not paint a complete picture of sustaining efforts for PI. For instance, case studies derive rich qualitative insights but have some limitations for empirical generalization when drawing inferences (Eisenhardt, 1989). Similarly, comparing pre- and post-implementation data alone does not account for unobservable factors that may affect outcomes and may not be able to explain comprehensively why certain efforts for PI succeed while others fail. Our research began inductively, with our intervention targeting the ill-structured problem of initiating, executing, and sustaining PI. We found that existing approaches derived from the literature required refinements and adjustments, which we developed with the participation of frontline employees. Next, using deductive logic, we examined the effects of the process redesign, controlling for specific unobservable factors such as EMR implementation and integration challenges pointed out as relevant by the caregivers. Our involvement also allowed us to access patient data from a control group of transplant patients from the same hospital that we used in our analyses. The IBR approach allowed us not only to identify the factors that promote sustaining PI but also to examine empirically the efficacy of the process redesign and PI through observations and analyses.

We find support for the effectiveness of the new redesigned process in improving both readmission rates and patient experience outcomes. Had we only collected pre- and post-redesign data from the kidney transplant patient education process, we would have incorrectly identified a performance deterioration in readmission outcomes in the course of the implementation. Our intervention-based quasi-experimental research design (Pagell & Shevchenko, 2014), which included data from the process undergoing redesign as well as data from a comparable process within the same organization, also overcomes some of the limitations of previous studies in deriving causal inferences.

6.2 | Implications for practice

Perhaps the most valuable practical insight gained from this research addresses critics who dismiss efforts for PI in healthcare as "medical Taylorism" (Hartzband & Groopman, 2016). Medicine is governed by the requirement to follow evidence-based standards, yet their implementation allows the care team much discretion. By seeking to eliminate process variation, the standardized work in this research somewhat limits that discretion but honors it nonetheless by introducing patient-centric elements (e.g., using multiple learning aids, having teach-back conversation, etc.). Applying standardized practices in repetitive processes has been shown to improve quality in many industries, yet processes are never "exactly repetitive" in a patient-care setting, because of the variation patients introduce. This, therefore, necessitates a standard process design that allows for patient-induced variation. Such design is necessarily iterative and the product of collaborative work by the care team, informed by needs the patient articulates. As such, it is not Taylorism. Rather than dictating scientific management, this standardization incorporates the ideas of those who do the work (including physicians and nurses) as well as those affected by it (patients). The design, implementation, and sustaining of standardized yet patient-centric processes requires intensive team effort. The evidence provided here suggests that it can result in greater patient satisfaction and better medical outcomes.

During the redesign, we observed the HCAHPS scores for the treatment group decline while 30-day readmissions remained unchanged. Upon discussing this result with the caregivers, a possible explanation emerged. The implementation required caregivers to dedicate time to workshops and training to change delivery of education for post-surgical care. Moreover, patients experienced caregivers' following now-standardized protocols more strictly. These two factors possibly drove lower ratings on the HCAHPS surveys. Supporting this possible explanation, multiple patients said in interviews conducted in the thick of the implementation period that they received instructions from a veteran nurse who was using the teach-back method with inpatient nurses, which degraded their patient experience. One patient even described feeling like a "test subject" at the time. Regardless of patients' perceptions, their health outcomes, measured using 30-day readmission rates, did not suffer in the implementation period. The deliberate and steady implementation of redesign with the active involvement of frontline employees at every stage enabled the employees to look out for external disruptions caused by the change and to navigate them.

These results point to two practical insights for implementing organizational initiatives for PI in high-interaction service processes. First, service providers must actively manage customers' perceptions of activities for PI when conducted in their presence to mitigate any potential decline in satisfaction. Second, measures of patient satisfaction and patient health outcomes may diverge during care delivery. It is important to note, however, that neither declined post-redesign. By striving for caregiver engagement in process redesign and including patient input, we also averted the expected initial dip in results following new-process adoption (Nembhard & Tucker, 2011; Song et al., 2017).

6.3 | Limitations and future research

We recognize the following limitations, which may serve as directions for future research. First, all insights from our study are derived from one hospital—Alpha. Although we collected data from multiple sources, validated our findings through post hoc employee and patient interviews, and ensured the patient mix in our sample compared with national demographics, we must exercise caution when generalizing our findings to other transplant units and hospitals. Second, we note that our empirical analyses are limited to the context of the patient education process after kidney transplant surgery, thus caution must be used in extending the benefits of such redesign to other types of healthcare delivery and to the sustainment of initiatives for PI in other high-interaction service contexts. Efforts are already under way at Alpha to adopt similar patient education process redesigns for its hip and knee replacement units. We encourage more research from other contexts, which can help strengthen the benefits of patient-centric standardized practices to healthcare delivery and sustainment of initiative for PI toward continuing process redesign efforts.

In conducting this research, we also came across three extensions that would be worth addressing in future research. First, our study documented a set of approaches to overcome challenges for sustaining PI. These approaches are not exhaustive, and more research on how healthcare organizations can combat these challenges would be fruitful. Second, we also see opportunities for research on the role of middle management. OM scholars have often discussed the role of senior management or frontline management, but very little is known on how middle managers affect PI in healthcare organizations. Third, although the role of handoffs has been touched on in the context of errors and safety in healthcare, their role in PI and the use of medical informatics for handoffs are areas that are worthy of examination.

7 | CONCLUSION

Notwithstanding the above limitations, our study offers important implications for managers and scholars. We provide detailed insights on how to trigger process redesign and PI, especially in the face of employee indifference and resistance. We demonstrate the key elements of employee engagement and customer input, and how they can enable standardization-adaptation in high interaction service settings. Finally, we also shed light on the often-ignored middle manager, essential for sustaining change and PI. Overall, our paper addresses the questions raised by skeptics about the use of standardized processes and PI in high-interaction service settings such as healthcare.

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ENDNOTES

[1] We did not include the unit in which heart transplant patients stayed, because that unit also had patients who had undergone other cardiac procedures, and as such, might not adequately reflect true HCAHPS scores related to transplants. Additionally, there were a few beds in the liver unit that were dedicated for heart transplant patients, given the similarities in post-transplant care and education.

[2] All kidney transplant recipients stayed in Unit A, while all liver and some heart transplants patients stayed in Unit B after surgery.

[3] All the heart and liver transplants were cadaveric organs

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APPENDIX A: DISCUSSION PROMPTS FOR FOCUS GROUPS

Describe the process of by which you received information and training for your post-transplant care

Explain some information received about the care after surgery?

Who delivered this information and training?

Did anyone contact you from the transplant unit after discharge? If yes, who contacted you and what did you discuss during the first call?

What about the training process for post-surgical care did you find to be particularly useful?

What about the training process for post-surgical care would you like to see changed?

Reflect on the first month after transplant. What would have been useful during the time from discharge instructions perspective?

APPENDIX B: LEARNING STYLES ASSESSMENT SURVEY ITEMS (KOLB, 1981)

All items scored on 5-point scale: (1) Strongly disagree, (2) Disagree, (3) Neither agree nor disagree, (4) Agree, (5) Strongly agree.

I find it easy to meet new people and make new friends

I am cautious and thoughtful

I get bored easily

I am a practical, "hands on" kind of person

I like to try things out for myself

My friends consider me to be a good listener

I have clear ideas about the best way to do things

I enjoy being the center of attention

I am a bit of a daydreamer

I keep a list of things to do

I like to experiment to find the best way to do things

I prefer to think things out logically

I like to concentrate on one thing at a time

People sometimes think of me as shy and quiet

I am a bit of a perfectionist

I am enthusiastic about life

I would rather "get on with the job" than keep talking about it

I often notice things that other people miss

I act first then think about the consequences later

I like to have everything in its "proper place"

I ask lots of questions

I like to think things through before getting involved

I enjoy trying out new things

I like the challenge of having a problem to solve

APPENDIX C: VARIABLES USED IN ANALYSIS

30-Day readmission: This variable indicates whether the patient was readmitted to this or any other hospital within 30 days after being discharged from this hospital.

Overall patient satisfaction: We used unit-level HCAHPS scores (see hcapsonline.org for details), which assess patient satisfaction with hospital stay on a scale of 0–10 (10 being the best). Consistent with previous research (Chandrasekaran et al., 2012; Senot et al. 2016b), we used the percentage of patients who gave scores of eight and greater, and adjusted them for factors such as education, self-rated health, primary language, age, socio-economic status, and delay in survey response. We obtained these scores for the treatment and control group units for each month in our study period.[2]

Implementation periods: Our study focused on differences in readmissions and overall HCAHPS performance between the treatment group (kidney) and control group (heart and liver) across three periods. The pre-redesign period captures all transplants performed between January 2013 and June 2014. In this period, there were 257 kidney, 44 liver, and 17 heart transplants. Next, *Redesign* is a binary variable representing transplants between July 2014 and April 2015. It is used to test the performance differences between the pre-redesign and redesign periods. There were 147 kidney, 15 heart, and 22 liver transplants conducted at the hospital in this period. Finally, *Post* is a binary variable representing transplants between May 2015 and April 2016. It is used to test the performance differences between the pre-redesign and post-redesign periods. There were 167 kidney, 37 liver, and 30 heart transplants in this period. From the

736 patients in the total original sample, we deleted 34 patients who had multiple transplants due to rejections that occurred in the course of their post-surgery hospital stay, and which lasted for more than 30 days after the initial procedure. Thus, the final sample for analyses included 702 patients (563 kidney, 48 heart, and 91 liver).

Patient controls: Our analyses controlled for several patient-level factors that can potentially impact readmissions and HCAHPS scores. They include patient age, length of stay, gender, type of transplanted organ[3] (living vs. cadaveric), and patient ethnicity (African American, Caucasian, and others). We also controlled for pre-existing conditions such as diabetes, discharge conditions such as transplanted organ function, and 30-day mortality. Finally, we added dummy variables to control for the year and quarter of transplant for readmission analyses, for year and month for the HCAHPS analyses.

Process controls: We controlled for factors reflecting other changes that occurred at the hospital through the period of our redesign and that may impact the performance of the patient education processes. Value-based purchasing (VBP) is a binary variable that captures the effect of the October 2013-launched program, which can affect the performance of the treatment and the control processes. In addition, we included the following three factors: electronic medical records implementation (EMR) that put caregivers through additional training, outpatient EMR glitches and integration issues at the outpatient facilities for both groups (Integration), and transition of care for the kidney unit (Transition), reflecting the June to August 2015 period when the outpatient clinic moved between facilities. We also checked the turnover levels for nurses, physicians, and surgeons, and found them to be steady throughout the period of our study, so we did not include them in our analyses. For the HCAHPS analysis, we controlled for the number of transplants conducted by the corresponding units, since a larger number of transplants in each month could affect the participation rates in the HCAHPS surveys.

APPENDIX D

Descriptive statistics

Label	Mean	*SD*	Definition
Dependent variables			
Readmit	0.37	0.48	Patient readmitted back within 30-days (1 = yes, 0 = no)
HCAHPS	0.69	0.10	Monthly overall patient satisfaction ratings for treatment and control groups
Independent variables			
Redesign	0.25	0.43	Binary variable to capture the process redesign period (July 2014 – April 2015, 0 = otherwise)
Post-redesign	0.16	0.37	Binary variable to capture the post-redesign period (May 2015 – April 2016, 0 = otherwise)
Patient controls			
Mortality	0.01	0.10	Binary variable to account for patient death within 30 days after transplant
LOS	8.87	4.82	Length of hospital stay of the patient (in days)
Age	51.5	12.3	Age of the transplant patient
Gender	0.38	0.48	Gender of the transplant patient (0 = male, 1 = female)
Kidney	0.81	0.39	Binary variable to capture if the patient had a kidney transplant
Heart	0.07	0.25	Binary variable to capture if the patient had a heart transplant
Donor type	0.37	0.49	Type of transplant donor (1 = living, 0 = cadaveric)
White	0.75	0.45	Binary variable capturing patient ethnicity (white); other is control group
African American	0.20	0.40	Binary variable capturing patient ethnicity (African American); other is control group
Graft health	0.09	0.28	Condition of the transplanted organ at discharge (1 = delayed function, 0 = healthy)
Pre-existing Cond	0.25	0.75	Patient with pre-existing diabetes condition (1 = yes, 0 = no)
Year dummies			Three-year dummies to reflect year of transplant (2013, 2014, 2015:2016—Base group)

(Continues)

(Continued)

Label	Mean	*SD*	Definition
Quarter dummies			Three quarter dummies to reflect the quarter in which transplant is done (Q1, Q2, Q3: Q4—Base quarter)
Month dummies			Eleven-month dummies to reflect the month of transplant for overall satisfaction rating analysis
Process controls			
VBP	0.81	0.39	Binary variable for the value-based purchasing program (1 = after Q3 2013, 0 = otherwise)
Transition	0.18	0.38	Binary variable for the transition of outpatient clinic for kidney (1 = June–August 2015, 0 = otherwise)
Integration	0.06	0.23	Binary variable to capture outpatient EMR issues for all transplants (1 = August 2015, 0 = otherwise)
# of transplants			Number of transplants performed by the treatment and control group per month

Note: For binary variables, the values reported indicate the proportion of patients undergoing transplant when the binary variable assumes a value of 1. For example, a mean of 0.81 for VBP indicates that 81% of the patients in the study sample underwent transplant surgery post-VBP.

APPENDIX E: CORRELATIONS

		1	2	3	4	5	6	7	8	9	10	11	12
1	Readmit	1											
2	HCAHPS	−0.06*	1										
3	Mortality	−0.03	0.01	1									
4	LOS	0.04	−0.17*	0.03	1								
5	Graft health	0.11*	0.06*	0.07*	0.08*	1							
6	Pre-existing condition	0.00	−0.04	0.04	0.02	−0.00	1						
7	Gender	0.09	0.01	0.02	−0.08	−0.01	−0.03	1					
8	Age	0.02	−0.07*	0.06*	0.09*	0.00	0.12*	−0.10*	1				
9	VBP	0.04	0.16*	0.01	−0.00	0.02	0.08*	−0.01	−0.01	1			
10	Donor type	−0.09*	0.16*	−0.05	−0.22*	−0.19*	−0.01	0.07*	−0.11*	0.00	1		
11	Transition	−0.01	0.15*	−0.04	−0.06*	0.08*	0.08*	0.05	−0.05	0.31*	−0.01	1	
12	Integration	0.06*	−0.18*	0.01	0.04	0.03	0.10*	0.00	0.00	0.67*	−0.06*	0.45*	1

*$p < 0.05$.

APPENDIX F: CHECKS FOR APPLICABILITY OF DID ANALYSIS

We checked for the three necessary conditions for applicability of DID analysis:

1. *Stable unit treatment value (SUTVA)*: SUTVA consists of two components: (a) the treatment applied does not affect the control group, and (b) there is a single version of the treatment. As to the first component, the treatment and control units were in different buildings in the hospital and did not have common caregivers. This reduced the possibility of carry-over learning effects between the two groups and ensured that the treatment applied in the form of the process redesign did not affect the control group. As to the second component, the intervention was designed such that the redesigned process of patient education was implemented uniformly. These aspects of our quasi-experiment design jointly support the two components of SUTVA.
2. *Parallel trends*: This condition requires that in the absence of treatment, the difference between the "treatment" and "control" group is constant over

time. Violation of this condition may result in biased estimation of the causal effect. We used two approaches to examine whether readmission rates and HCAHPS scores between kidney and other transplant groups exhibited a parallel trend prior to our intervention. First, we created a trend variable and included an interaction between the trend and the treatment variable (i.e., kidney vs. other transplants) in the regression model. This regression model was then fit for the *pre-redesign period*. The interaction term was used to compare the trend in readmissions and HCAHPS between the treatment and control groups for the *pre-redesign period*. The coefficients for the interaction term were nonsignificant, thus satisfying the parallel trends condition. Second, we used the approach outlined in Pischke (2005). We included interactions between the time dummies and treatment indicator (i.e., kidney vs. other transplants) for all three periods, with the first *pre-redesign period* for the control group as the baseline. The interaction coefficients were nonsignificant, providing further evidence of satisfying the parallel trends condition.

3. *Strict exogeneity*: This condition implies that any treatment exposures in the *post-redesign period* are not anticipated by outcomes in earlier periods, that is, there were no changes in the patient education process in anticipation of the treatment rollout. In our setting, the rollout of the new patient education process and supporting material was carefully timed for the execution phase. This ensured compliance with the strict exogeneity condition.